



Predictive model for estimating liquid film thickness in upward vertical annular gas–liquid flow: a semi-empirical approach for energy-related applications

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Abstract

The liquid film thickness in vertical annular gas–liquid two-phase flow is a critical parameter in the design and analysis of thermal systems used in power generation and other engineering applications, including heat exchangers and related two-phase flow equipment. This study presents a new semi-empirical correlation developed based on the Henstock and Hanratty framework, utilizing a comprehensive database consisting of 876 experimental data points collected from nine independent data sources. The correlation parameters were determined using a highly reliable Source-weighted Nonlinear Least-Squares Regression approach, enabling applicability over a broad range of operating conditions. To address overfitting related to the liquid-to-gas viscosity ratio, the proposed model incorporates three principal dimensionless groups: the gas Reynolds number, the gas-to-liquid mass flux ratio, and the gas-to-liquid density ratio. The resulting correlation exhibits robust predictive performance, surpassing existing correlations in the literature, with a mean absolute percentage error (MAPE) of 21.34% and a coefficient of determination (R^2) of 0.8939. Additionally, 77.51% of the predicted values are contained within a $\pm 30\%$ error band. Although the proposed model is expressed in an explicit closed-form equation, making it convenient for practical engineering calculations, certain limitations remain, particularly for applications involving non-water and high-viscosity fluids. These aspects warrant further investigation and future refinement. Overall, this study provides a reliable and well-validated predictive tool for the design and optimization of heat-transfer systems involving annular gas–liquid two-phase flow across a wide range of engineering applications.

Keywords Annular flow · Liquid film thickness · Gas–liquid flow · Two-phase flow · Semi-empirical models

Abbreviations

D	Pipe diameter [m]
F	Flow parameter [–]
F^*	Modified flow parameter [–]
Fr	Froude number [–]

f	Friction factor [–]
g	Acceleration due to gravity [m s^{-2}]
j	Superficial velocity [m s^{-1}]
Re	Reynolds number [–]
u^*	Friction velocity [m s^{-1}]
We	Weber number [–]
x	Quality [–]

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Greeks

δ	Liquid film thickness [m]
μ	Dynamic viscosity [N s m^{-2}]
ν	Kinematic viscosity [$\text{m}^2 \text{s}^{-1}$]
ρ	Density [kg m^{-3}]
σ	Surface tension [N m^{-1}]
τ	Shear stress [N m^{-2}]

Subscripts

C	Characteristic
f	Liquid phase
g	Gas phase

i Interfacial
s Surface
w Wall

Introduction

The two-phase annular flow pattern is commonly found in various industrial applications, including those in the petroleum and chemical industries, as well as boiling and condensing heat-transfer equipment, such as condensers and evaporators [1, 2]. Two-phase flow produces significantly higher pressure drops than single-phase flow because both gas and liquid must share the same flow space within a tube. However, its convective heat-transfer coefficient is substantially greater, making it suitable for high-heat-flux thermal management applications [3].

Annular flow consists of a fast-moving gas core surrounded by a slower-moving liquid annular film along the pipe wall, with entrained liquid droplets in the gas core [4]. A high gas flow rate induces strong shear forces at the gas–liquid interface, sometimes leading to formation of a wavy interfacial structure [5, 6]. Liquid droplets in the gas core undergo continuous entrainment from the liquid film and deposition back onto it. At a steady state, the rates of droplet entrainment and deposition reach equilibrium, establishing a stable entrained droplet fraction within the gas core [7].

One of the most significant aspects of annular flow is its impact on convective heat transfer. Heat transfer is enhanced due to the large velocity and temperature gradients within the thin liquid film near the wall [1]. These gradients can be further augmented by either increasing the gas flow rate or decreasing the liquid film thickness [3]. However, excessive thinning of the liquid film can lead to film rupture, causing dry-out conditions. This can result in catastrophic equipment failure due to greatly increased wall temperatures [8].

It is essential to develop accurate predictive models for liquid film thickness to effectively utilize annular flow in thermal management applications, such as energy systems and advanced cooling mechanisms. Recent studies have introduced empirical correlations and numerical models to improve the precision of liquid film thickness predictions [1, 9]. Additionally, two-fluid modeling has proven effective in predicting pressure gradients, interfacial friction, and film characteristics under varying viscosity [4]. Advanced experimental investigations on disturbance waves and liquid entrainment mechanisms have provided deeper insights into annular flow stability, highlighting the interplay between interfacial shear stress and wave dynamics [5, 6, 9]. Accurate liquid film thickness prediction remains a fundamental requirement to practically and efficiently integrate annular flow into high-heat-flux thermal management applications.

Predictive models of the liquid film thicknesses of annular flows have been developed. They can be categorized into mechanistic and empirical approaches. Mechanistic approaches derive mathematical formulations that establish relationships among important flow parameters based on fundamental physical principles. These models can be verified and improved with limited experimental data. However, since mechanistic approaches are grounded in physical phenomena, they can be valid across a wide range of operating conditions. A commonly used mechanistic model for annular flow was developed by Alves et al. [10]. The major drawback of mechanistic models is that their flow parameters are implicitly expressed and coupled, making implementation quite complicated in many cases.

Empirical approaches are alternative methods for evaluating the liquid film thickness of an annular flow [7, 11–15]. Numerous correlations have been developed using this approach. An empirical approach primarily involves using experimental data for curve fitting to establish relationships between normalized film thickness and other related non-dimensional parameters. However, empirical correlations lack theoretical grounding, limiting their applicability outside the experimental range from which they are derived. They offer poor physical interpretability and are prone to overfitting, reducing their reliability when new or varying conditions are encountered. Semi-empirical correlations offer several advantages over purely empirical models because they combine theoretical principles with experimental data. As a result, they generally provide predictions that are more physically meaningful and applicable over a wider range of operating conditions. At the same time, they help reduce the risk of overfitting to specific datasets. A semi-empirical approach, sometimes considered a subcategory of the empirical approach, involves theoretically simplifying general physical relations to obtain a functional form [16–18]. Most studies tend to adopt either fully empirical approaches based purely on experimental data, or develop complex analytical models grounded entirely in theoretical formulations. Semi-empirical correlations have received relatively limited attention in the literature.

Although several semi-empirical correlations for liquid film thickness have already been proposed, significant research gaps still remain, particularly regarding the quantity and diversity of the experimental data used in developing those correlations. To ensure the robustness and general applicability of a model, the use of a large and comprehensive dataset is essential. Additionally, attention should be given to the practicality and ease of implementation of these correlations in real world engineering applications.

To address this limitation, the current study develops a semi-empirical model that utilizes a large pool of experimental data, mitigating unintentional bias and improving both its reliability and accuracy. The developed

correlation, which has not previously appeared in the open literature, is presented in a closed-form expression with a simple mathematical structure, making it straightforward and convenient to apply. This contribution provides a robust and validated predictive tool for enhancing the design and optimization of efficient and effective two-phase annular flow systems.

Although the dimensionless groups considered in this study namely the gas Reynolds number, the gas-to-liquid mass flux ratio, the viscosity ratio, and the density ratio have individually been applied in prior research, the novelty of this work lies in their integration into a unified, semi-empirical flow parameter supported by a significantly larger and more diverse experimental database. Unlike prior correlations, which were often tailored to limited experimental conditions or relied heavily on implicit mechanistic formulations, the present correlation provides an explicit closed-form expression that is straightforward to implement. Furthermore, the compiled database of 876 data points from nine independent sources represents one of the most comprehensive datasets available for vertical annular flow. This combination of a broad database and an explicit correlation framework address the limitations of earlier studies, which either lacked generalizability or suffered from overfitting to narrow experimental conditions. The proposed approach therefore provides not only improved predictive accuracy but also enhanced applicability in engineering practice, especially for energy-related systems such as nuclear cooling and advanced heat exchangers.

Methodology for data collection and verification

Data collection

The database used in the current study is created from available data on liquid film thickness in annular vertical two-phase flow that is published, accessible, retrievable, and contains sufficient information for repeatability. The experimental data is extracted from graphs or tables (where available) into data points. The collected liquid film thickness data is compiled from nine different experimental studies, resulting in a database of 876 data points that is summarized in Table 1. The details of the data are as follows.

Bousman and McQuillen [19] experimentally studied air–liquid annular flow in a 12.5-mm diameter tube under microgravity conditions. Three liquids were used in the experiment to investigate the effects of surface tension and viscosity on film thickness. Acrylic tubes equipped with parallel wire film thickness conductance probes were used to measure film thickness. Their results showed that film thickness decreases with greater air and lower water velocities. The film thickness in air–water/glycerin and air–water/Zonyl was 20–30% and 40–50% greater than that of the air–water experiments, respectively.

Fukano and Furukawa [12] studied the effects of liquid viscosity on the film thickness of vertical air–liquid annular flow in a 26-mm diameter tube using water and aqueous glycerol solutions as liquid phases. Film thickness was measured using a constant electric current method, developed by these researchers. Their results showed that, at similar

Table 1 Summary of collected data for liquid film thicknesses of annular flow in a vertical direction

Author	Fluid	D/mm	$j_l/\text{m s}^{-1}$	$j_g/\text{m s}^{-1}$	P/bar	Number of data points
Bousman and McQuillen [19]	Air–water <i>Microgravity</i> ($0 \pm 0.02g$)	12.70	0.07–0.5	4–26	Ambient	26
Fukano and Furukawa [12]	<i>Air–water</i>	26.00	0.04–0.1	10–50	1.03–1.17	17
	<i>Air– 45 mass%/glycerol</i>	26.00	0.04–0.1	10–50	1.03–1.17	18
	<i>Air– 53 mass%/glycerol</i>	26.00	0.04–0.1	10–50	1.03–1.17	18
	<i>Air– 60 mass%/glycerol</i>	26.00	0.04–0.1	10–50	1.03–1.17	18
Gill et al. [20]	Air–water	31.75	0.004–0.198	13.44–80.65	Ambient	58
MacGillivray [13]	<i>Air–water</i>	9.50	0.076–0.315	13.0–29.4	0.74–1.37	221
	Helium–water	9.50	0.098–0.312	22.2–62.4	0.92–1.73	108
Sawant [21]	Air–water	9.40	0.050–0.542	2.44–98.3	1.04–6.07	157
Ashwood et al. [22]	Air–water	23.40	0.063–0.338	32.57–77.95	Ambient	46
Schubring et al. [23]	Air–water	23.40	0.040–0.340	32.80–114.30	Ambient	122
Alamu [24]	Air–water/glycerol	19.00	0.050–0.150	13.82–42.90	1.4	37
Wang et al. [25]	Air–water	25.4	0.061–2.049	8.82–26.46	1.07–1.37	30
				Total data point	876	

Reynolds numbers, the liquid film in air–water flow was thinner than that in air–glycerol flow, indicating that film thickness increases with greater dynamic liquid viscosity.

Ashwood et al. [22] investigated air–water flows in vertical annular flow in a 23.4-mm diameter tube at room temperature and atmospheric pressure. They directly measured film thickness using total internal reflection (TIR) and planar laser-induced fluorescence (PLIF). These techniques produced similar results. Later, Schubring et al. [23] provided a more detailed explanation of the PLIF technique and applied it to gather additional data on the film thickness of vertical air–water annular flows. Results from both experiments indicated that film thickness decreases with greater air velocity and decreasing water velocity.

Sawant [21] conducted experiments on air–water and an organic fluid (Freon-113) with vertical annular flow in a 9.4-mm diameter tube under pressures ranging from 1.2–6 bar and 3–8.5 bar, respectively. The study found that greater pressures result in higher air densities, leading to decreased liquid film thicknesses. Notably, the current work utilizes the comprehensive revised dataset provided by Ju et al. [15], who identified and corrected errors in the original calibration curves used in the study by Sawant [21]. MacGillivray [13] studied air–water and helium–water vertical annular flow under ambient pressure and temperature in a 9.5-mm diameter tube. The study considered the impacts of gas density and gravity as factors influencing liquid film thickness.

Wang et al. [25] conducted an experimental study using air–water as fluids at pressures ranging from 107 to 137 kPa to understand wave characteristics in high liquid flow rate annular flow, known as wispy annular flow. They used conductance film thickness probes to characterize the film structure of upward annular flow at high temporal resolution. In the present work, the superficial velocities and properties were extracted at the measurement location of $z/D_h = 141$. This ensures that the data represents the fully developed flow conditions, as the study identified a development length of $z/D_h \approx 85$ for their experimental setup.

A verification process was conducted to ensure the accuracy of the collected liquid film thickness database. Each dataset compiled from the same source was compared with the predicted results of the corresponding model. This comparison aimed to assess the consistency between the collected data and the originally reported values. The accuracy of this verification was evaluated using the mean absolute percentage error (MAPE) of each model, defined as

$$\text{MAPE} = \frac{1}{N} \sum_i^N \left| \frac{\delta_{\text{pred},i} - \delta_{\text{exp},i}}{\delta_{\text{exp},i}} \right| \quad (1)$$

against the reported mean absolute percentage error in the literature. This shows that the data has been compiled correctly.

Published predictive models

After creating the database and verifying its accuracy, its experimental data were used to evaluate the accuracy and reliability of the published models. Table 2 summarizes some of the correlations reported in the literature for predicting the liquid film thickness in vertical annular flow.

Henstock and Hanratty [16] related the normalized liquid film thickness to the ratio of interfacial to gas-phase friction factors to derive an equation for normalized liquid film thickness. Tatterson et al. [17] later modified this model to adjust the influence of the liquid Reynolds number. Based on their experimental data, Hori et al. [11] considered the Reynolds and Froude numbers of both liquid and gas phases, along with the ratio of the liquid-to-water dynamic viscosities at a 20 °C reference temperature. These served as governing parameters for their correlation.

Fukano and Furukawa [12] introduced a trial and error correlation to predict normalized film thickness as a function of the gas Froude number, liquid Reynolds number, and quality, based on their experimental data. de Jong [26] performed a dimensional analysis on microgravity annular flow data. They proposed a correlation for the film Reynolds number as a function of liquid Reynolds number, quality, and the ratio of gas to liquid density. MacGillivray [13] later modified the empirical constant in de Jong's correlation based on his experimental data.

Berna et al. [14] reviewed existing relationships and alternatively developed their correlation as a function of the Reynolds and Froude numbers of both liquid and gas phases, using published experimental data. Ju et al. [15] postulated that the non-dimensional parameters used in prior correlations either rely on variables that are difficult to accurately determine (e.g., interfacial shear stress) or fail to adequately represent the physical phenomena. They suggested that the correlation for film thickness should be expressed in terms of film thickness normalized by the maximum film thickness rather than tube diameter. The maximum film thickness can be determined from the slug-to-churn flow transition criteria developed by Mishima and Ishii [27]. Their proposed empirical correlation, expressed as a function of the liquid Weber number, modified gas Weber number, and non-dimensional viscosity number, takes the form of a hyperbolic tangent function, ensuring that the range remains limited between 0 and 1.

Ju and Hibiki [18] developed a semi-empirical correlation for dimensionless liquid film thickness. Their general expression, derived from a one-dimensional force balance analysis of the liquid film in annular flow, emphasizes the interplay among wall shear stress, interfacial shear stress, and gravitational forces. Initially, after curve fitting with experimental data, they proposed the dimensionless liquid film thickness in relation to the interfacial Reynolds

Table 2 Summary of correlations for liquid film thicknesses in annular flow

References	Correlation
Henstock and Hanratty [16]	Vertical flow $\frac{\delta}{D} = \frac{6.59F}{(1+1400F)^{0.5}}$ Horizontal flow $\frac{\delta}{D} = \frac{6.59F}{(1+850F)^{0.5}}$ where $F = \frac{1}{\sqrt{2}} \frac{\text{Re}_g^{0.5} \mu_f \rho_g^{0.5}}{\text{Re}_g^{0.5} \mu_g \rho_f^{0.5}}$
Tatterson et al. [17]	Vertical flow $\frac{\delta}{D} = \frac{6.59F}{(1+1400F)^{0.5}}$ Horizontal flow $\frac{\delta}{D} = \frac{6.59F}{(1+850F)^{0.5}}$ where $F = \frac{\gamma(\text{Re}_f)}{\text{Re}_g^{0.9} \mu_g \rho_f^{0.5}}$ and $\gamma(\text{Re}_f) = [(0.707\text{Re}_f^{0.5})^{2.5} + (0.0379\text{Re}_f^{0.9})^{2.5}]^{0.4}$
Hori et al. [11]	$\frac{\delta}{D} = 0.905\text{Re}_g^{-1.45}\text{Re}_f^{0.9}\text{Fr}_g^{0.93}\text{Fr}_f^{-0.68}\left(\frac{\mu_f}{\mu_{\text{water}@20^\circ\text{C}}}\right)^{1.06}$
Fukano and Furukawa [12]	$\frac{\delta}{D} = 0.0594\exp(-0.34\text{Fr}_g^{0.25}\text{Re}_f^{0.19}x^{0.6})$ where $x = \frac{(j_g)\rho_g}{(j_g)\rho_g + (j_f)\rho_f}$
MacGillivray [13]	$\frac{\rho_f(j_f)\delta}{\mu_f} = 39\text{Re}_f^{0.2}\left(\frac{1-x}{x}\right)\left(\frac{\rho_g}{\rho_f}\right)^{0.5}$
Berna et al. [14]	$\frac{\delta}{D} = 7.165\text{Re}_g^{-1.07}\text{Re}_f^{0.48}\left(\frac{\text{Fr}_g}{\text{Fr}_f}\right)^{0.24}$
Ju et al. [15]	$\frac{\delta}{D} = 0.071 \tanh\left(14.22\text{We}_f^{0.24}\text{We}_g'^{-0.47}N_\mu^{0.21}\right)$ where $\text{We}_g' = \frac{\rho_g(j_g)^2 D}{\sigma} \left(\frac{\Delta\rho}{\rho_g}\right)^{1/4}$ and $N_\mu = \frac{\mu_f}{\sqrt{\rho_f \sigma} \sqrt{\frac{\sigma}{g\Delta\rho}}}$
Ju and Hibiki [18]	$\frac{\delta}{D} = 0.797\left(\frac{\rho_f(j_f)^2}{\rho_g(j_g)^2}\right)^{0.282}\left(\frac{D}{\sqrt{\sigma/\Delta\rho}}\right)^{-0.296}$

number ratio, the interfacial inertia ratio, and a dimensionless pipe diameter. They later simplified this correlation by neglecting the interfacial Reynolds number ratio. However, because the datasets used to develop their correlation did not include fluids with varying viscosities and surface tensions, the effects of these properties were not evaluated in their correlation.

To provide a more comprehensive and robust assessment of the existing literature, multiple statistical indicators are employed to quantify the predictive accuracy of published models against the experimental database. Table 3 summarizes the global performance metrics, including the root mean square error (RMSE), bias (systematic error), coefficient of determination (R^2), mean

Table 3 Global statistical performance comparison of predictive models

Correlation	RMSE $\times 10^3$ /—	Bias $\times 10^{4**}$ /—	R^2 /—	MAPE / %	Within $\pm 30\%$ / %
Henstock and Hanratty [16]	9.73	− 47.36	0.4799	39.17	45.09
Tatterson et al. [17]	7.96	− 15.44	0.6525	47.33	57.88
Hori et al. [11]	234.55	923.95	− 301.0397	403.86	5.02
Fukano and Furukawa [12]	8.92	− 49.55	0.5627	33.29	48.52
MacGillivray [13]	10.89	− 0.70	0.3492	37.88	65.75
Berna et al. [14]	45.52	150.51	− 10.3746	86.73	38.93
Ju et al. [15]	5.60	− 17.77	0.8276	26.32	70.09
Ju and Hibiki [18]	7.46	7.35	0.6944	39.35	62.10
Developed Correlation*	4.40	8.80	0.8939	21.34	77.51

*The developed correlation in Sect. "Development of a new correlation"

**Positive value = overpredict

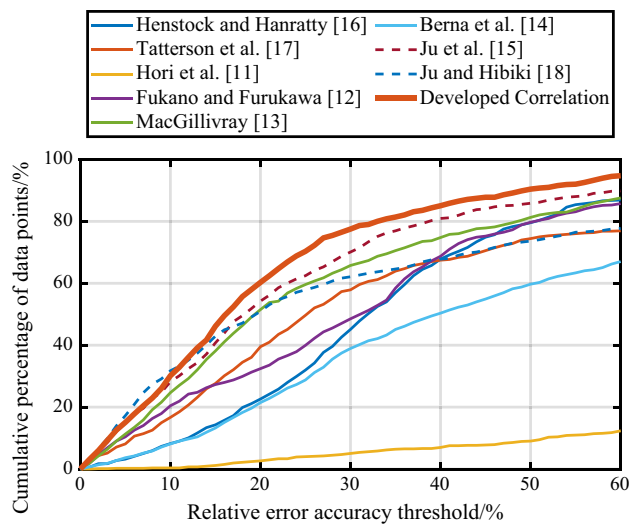


Fig. 1 Cumulative distribution of prediction accuracy across experimental datasets

The applicability of these models across a wide range of conditions is further illustrated in Fig. 1, which displays the cumulative percentage of data points predicted within a specified relative error accuracy threshold. This parameter illustrates how well a model can be applied over a wide range of conditions if some error is allowed. Generally, a smaller MAPE leads to a larger percentage of data points within the specified error; however, this is not always the case. For example, as shown in Table 3 and Fig. 1, while the model of Fukano and Furukawa [12] has a smaller MAPE than that of MacGillivray [13], the latter model achieves

a significantly higher percentage of data points within the $\pm 30\%$ error threshold (65.75% compared to 48.52%).

Comparison of results clearly shows that empirical models generally best agree with the set of data upon which they were developed. Table 5 summarizes the relationship between the predictive models and the data used for their development. Ju et al. [15] developed their model using experimental data from Sawant [21] and Fukano and Furukawa [12]. Table 4 shows that their model produces the smallest MAPE when compared with data from Sawant [21] and Fukano and Furukawa [12]. This trend is also observed in the model developed by MacGillivray [13]. Additionally, since this composite database includes 329 data points from MacGillivray [13], which constitute a large portion of the compiled database, an overall small error results with a large percentage of data points within the specified error bounds.

Generally, empirical and semi-empirical models developed using a single dataset are often not applicable to other experiments. However, it is notable that Fukano and Furukawa's [12] work produced the only model in this study that was developed using data solely from their own experiments. It exhibits a relatively small overall MAPE and a high overall percentage of data points within the specified error. This model even performs better than other models that use multiple datasets for model development. This is because they conducted experiments specifically to study the effects of liquid viscosity on film thickness, using four separate liquid samples with different viscosities. Their dataset can be considered as four separate sets of experimental data covering a wide range of operating conditions. This emphasizes that liquid film thickness is highly dependent on liquid properties. It highlights the

Table 5 Summary of data for published correlations of predicted liquid film thickness

Correlation	Source of experimental data	
	Vertical data	Horizontal data
Henstock and Hanratty [16]	Willis [28] Whalley et al. [29] Gill et al. [20] Chien and Ibele [30] Charvonia [31]	Butterworth [32] Swanson [33]
Tatterson et al. [17]	—	Tatterson et al. [17]
Hori et al. [11]	Hori et al. [11]	—
Fukano and Furukawa [12]	Fukano and Furukawa [12]	—
MacGillivray [13]	MacGillivray [13]	—
Berna et al. [14]	Cousins [34] Alamu [24] Schubring et al. [23]	Tatterson et al. [17] Paras and Karabela [35] Schubring and Shedd [36]
Ju et al. [15]	Sawant [21] Whalley et al. [29] Fukano and Furukawa [12]	
Ju and Hibiki [18]	Sawant [21] Whalley et al. [29] Wang et al. [25]	

importance of having a comprehensive database of experimental data covering a broad range of operating conditions for development of an accurate and widely applicable predictive model.

Development of a new correlation

Physical motivation and variable selection

This work develops a new model following the approach proposed by Henstock and Hanratty [16]. Their model is explicit and semi-empirical. It is expected to be applicable over a wider range of operating conditions. They formulated a liquid film thickness model by considering the shear stress equation at the fully developed wall layer in a vertical tube. The approach employs a force balance and relates the shear stress distribution in a liquid film to the velocity gradient using the eddy viscosity concept. Then, they solved the equations for the dimensionless film thickness. Ultimately, with the assumption that the dimensionless film thickness is relatively small, it is expressed as a function of the Reynolds number of the liquid film as

$$\delta^+ = \gamma(\text{Re}_f) \quad (2)$$

where δ^+ is the dimensionless film thickness. It can be defined as

$$\delta^+ = \frac{\delta u^*}{\nu_f} \quad (3)$$

where u^* is the friction velocity and ν_f is kinematic viscosity of the liquid film. The friction velocity can be computed from

$$u^* = \sqrt{\frac{\tau_c}{\rho_f}} \quad (4)$$

where τ_c is characteristic shear stress and ρ_f is density of the liquid film. Then the characteristic shear stress can be computed from

$$\tau_c = \frac{2}{3}\tau_w + \frac{1}{3}\tau_i \quad (5)$$

where τ_w is the wall shear stress and τ_i is the interfacial shear stress.

With the correlation for the surface friction factor on a smooth surface, the friction factor is

$$f_s = 0.049\text{Re}_g^{-0.2}. \quad (6)$$

Then, Eq. (2) can be rewritten in terms of δ/D as

$$\frac{\delta}{D} = \frac{6.59F\sqrt{\tau_i/\tau_c}}{\sqrt{f_i/f_s}} \quad (7)$$

where f_i is the interfacial friction factor and F was defined by Henstock and Hanratty [16] as a flow parameter such that

$$F = \frac{\gamma(\text{Re}_f)}{\text{Re}_g^{0.9}} \frac{\nu_f}{\nu_g} \sqrt{\frac{\rho_f}{\rho_g}} \quad (8)$$

For a relatively small dimensionless film thickness, the ratio of the interfacial shear stress to the characteristic shear stress, τ_i/τ_c , approaches one. The ratio of the interfacial friction factor to the surface friction factor, f_i/f_s , is only a function of the flow parameter, F

$$\frac{f_i}{f_s} = g(F) \quad (9)$$

Therefore, the functionality of the film thickness model proposed by Henstock and Hanratty [16] is

$$\frac{\delta}{D} = \frac{cF}{\sqrt{g(F)}} \quad (10)$$

where c is a coefficient. After this, they performed an empirical procedure to determine the final correlation shown in Table 2. A more detailed explanation of the model development can be found in the original paper [16].

The development of the proposed correlation is grounded in the fundamental governing mechanisms of interfacial friction and film thickness in upward annular flow. The original formulation by Henstock and Hanratty [16] established a semi-empirical model based on shear stress balance and turbulent eddy viscosity assumptions. However, their flow parameter was derived under restrictive conditions—specifically smooth pipe surfaces, air–water systems, and small interfacial shear stress ratios—which limited its generalizability to broader industrial operating conditions.

In the current work, the formulation was modified to address these limitations by explicitly incorporating four dimensionless ratios that characterize the dynamic balance and momentum transfer between phases: the gas Reynolds number (Re_g), the gas-to-liquid mass flux ratio ($x/(1-x)$), the density ratio (ρ_g/ρ_l) and the viscosity ratio (μ_g/μ_l). The proposed flow parameter, F^* , is defined as:

$$F^* = \text{Re}_g^{n_1} \left(\frac{x}{1-x} \right)^{n_2} \left(\frac{\rho_g}{\rho_l} \right)^{n_3} \left(\frac{\mu_g}{\mu_l} \right)^{n_4} \quad (11)$$

where x is vapor quality while n_1 , n_2 , n_3 and n_4 are exponents.

From a physical perspective, the gas-to-liquid mass flux ratio was selected over the traditional liquid Reynolds number due to its superior ability to represent interfacial

shear effects. Experimental sensitivities indicate that this ratio more effectively captures the momentum transfer from the gas core to the liquid film, which promotes droplet entrainment and subsequent film thinning. Furthermore, by removing the square-root dependency found in the original Henstock and Hanratty denominator, the new correlation provides greater empirical flexibility for non-air–water systems while maintaining physical interpretability. The resulting correlation for the normalized film thickness (δ/D) is expressed as:

$$\frac{\delta}{D} = \frac{c_1 F^*}{1 + c_2 F^*} \quad (12)$$

This structure ensures that the model retains a mechanistic foundation while improving prediction accuracy across a wide spectrum of fluid properties and operating ranges.

Optimization algorithm and constraints

To align this physical framework with statistical rigor, the coefficients and exponents of the proposed correlation were determined using a Source-weighted Nonlinear Least-Squares Regression. This methodology was specifically designed to mitigate the dominance of large datasets, such as the MacGillivray [13] source (constituting ~40% of the data points), by ensuring that the model development remains unbiased and equitable across all data sources.

Although the database consists of 876 data points compiled from nine independent experimental studies, the model development was expanded to 11 folds by treating distinct fluid systems within the same study as separate sources. Specifically, the experimental work of Fukano and Furukawa [12] was divided into two distinct sources based on the liquid phase: Air–water and Air–water/glycerol mixtures. Similarly, the study by MacGillivray [13] was split into Air–water and Helium–water systems to account for the drastic change in gas-phase density. By treating these as 11 independent sources, the LOSO-CV provides a more stringent test of the model's sensitivity to fluid property variations.

The optimization aimed to minimize the Weighted Residual Sum of Squares (RSS_w), defined as:

$$RSS_w = \sum_{i=1}^N w_i (\delta_{\text{exp},i} - \delta_{\text{pred},i})^2 \quad (13)$$

where the weighting factor w_i for each data point is defined as the inverse of the number of observations within its respective source (N_{source}): $w_i = 1/N_{\text{source}}$. This scheme effectively prevents "implicit overfitting" to any single dominant dataset and enhances the overall generalizability.

The regression utilized the Trust-Region-Reflective algorithm in MATLAB. This algorithm is particularly suited for solving large-scale constrained nonlinear problems. To

maintain physical consistency and ensure that the identified exponents reflect established fluid dynamic trends, the search space was restricted using the following constraints:

- *Coefficients* (c_1, c_2) Lower bound of 0 and upper bound of 10^5 to prevent non-physical negative values while providing sufficient flexibility for scaling.
- *Exponents* (n_1, n_2, n_3) Constrained within a range of $[-1, 1]$ to ensure the power-law relationships remain within physically plausible limits observed in interfacial shear and film flow literature.

To maintain neutrality and avoid local minima bias, the initial parameter values for the optimization were set as the midpoint of the specified bounds. High-precision convergence was ensured by defining strict stopping criteria, with both the function tolerance and step size tolerance set at 10^{-8} .

Leave-one-source-out cross-validation (LOSO-CV) and parameter sensitivity

To rigorously evaluate the model's generalizability and confirm that the weighting strategy successfully mitigated bias, a 11-fold LOSO-CV procedure was implemented. In each iteration, the model was trained on 10 sources and validated against the remaining unseen source. The validation performance is quantified by the average validation R_{cv}^2 :

$$R_{\text{cv}}^2 = 1 - \frac{\sum (\delta_{\text{exp,test}} - \delta_{\text{pred,test}})^2}{\sum (\delta_{\text{exp,test}} - \bar{\delta}_{\text{exp,test}})^2} \quad (14)$$

In a strict validation framework, the R_{cv}^2 serves as a direct indicator of the model's generalization power. Unlike the standard R^2 obtained from fitting all data, the R_{cv}^2 should ideally be as close to 1.0 as possible, but it must strictly be positive. A positive R_{cv}^2 indicates that the proposed correlation provides a better prediction of unseen data than a simple average of that data. Conversely, a negative R_{cv}^2 signifies that the model fails to capture the underlying physics of the independent dataset, often indicating overfitting to the training sources or a lack of physical robustness in the model's structure.

The stability of the identified parameters was assessed using Parameter Sensitivity Analysis by calculating the Coefficient of Variation (COV) for each exponent across the 11 LOSO-CV folds:

$$\text{COV}(\%) = \left(\sigma_p / |\mu_p| \right) \times 100 \quad (15)$$

where μ_p and σ_p are the mean and standard deviation of the parameter values ($c_1, c_2, n_1, n_2, n_3, n_4$) obtained from

each fold. In terms of parameter reliability, a low COV is an essential requirement. Typically, a COV below 10–20% confirms that the identified power-law relationships are fundamental and remain invariant regardless of the specific datasets used for training. High COV values would indicate that the parameters are highly sensitive to the training data, suggesting that the identified exponents are mere mathematical artifacts of a dominant dataset rather than a representation of the universal physical mechanisms governing film behavior.

Model refinement and dimensionless parameter selection

The selection of the final model was driven by a rigorous evaluation of predictive accuracy and parameter stability across a composite database characterized by significant physical diversity. Initially, a 4-variable formulation (Eq. 11) incorporating the liquid-to-gas viscosity ratio was tested to account for fluid property variations. However, as summarized in Table 6, despite achieving a slightly higher overall R^2 (0.9023), it exhibited a severe lack of generalizability. The LOSO-CV analysis for the 4-variable model yielded a negative average validation R_{cv}^2 (−0.3795) and extreme instability in the viscosity exponent (n_3), with a Coefficient of Variation (COV) reaching 1217.16%. This suggests that the inclusion of the viscosity ratio as a standalone exponent leads to severe overfitting, as its influence is either already indirectly captured by other parameters or insufficiently varied across the available databases.

Following the exclusion of the viscosity ratio, the refinement process investigated the optimal dimensionless groups to characterize the film behavior. Although the initial assumption prioritized the mass quality ratio ($x/(1-x)$), several studies in the literature emphasize the role of surface tension and gravity in interfacial wave formation and film stability, typically represented by the liquid Weber number (We_f) and liquid Froude number (Fr_f).

To address this, a comparative analysis was performed between three 3-variable model structures:

1. **3-Var-Quality-based (Proposed):** Based on Re_g , $x/(1-x)$, and ρ_g/ρ_f .
2. **3-Var-Weber-based:** Based on Re_g , We_f , and ρ_g/ρ_f .
3. **3-Var-Froude-based:** Based on Re_g , Fr_f , and ρ_g/ρ_f .

It is important to note that We_f and Fr_f were evaluated as direct replacements for the quality ratio $x/(1-x)$ rather than additional fourth variables. This strategic selection was made because both We_f (incorporating $\rho_f j_f^2 D/\sigma$) and Fr_f (incorporating $j_f/\sqrt{gD}$) share the liquid superficial velocity (j_f) as a core component, similar to the mass quality ratio $x/(1-x)$. Integrating these parameters simultaneously into a 4-variable structure would introduce high multi-collinearity and significant numerical instability during the nonlinear regression process. By maintaining a 3-variable framework, the model ensures physical robustness while identifying which liquid-phase dimensionless group provides the most accurate empirical fit across the diverse experimental conditions.

As detailed in Table 6, a comparative analysis of the 3-variable candidates reveals distinct differences in their predictive capabilities. The 3-Var-Weber-based model, while showing high numerical stability in its exponents (Max COV of 6.61%), provided a lower overall R^2 (0.8751) and a significantly higher MAPE (28.37%) compared to the quality-based approach. The 3-Var-Froude-based model performed the least effectively, yielding the highest MAPE (33.45%) and a weak average validation R_{cv}^2 of only 0.1073.

In contrast, the 3-Var-Quality-based model was selected as the optimal structure because it offers the superior balance between predictive accuracy and statistical robustness. It achieved the highest overall R^2 among all 3-variable models (0.8939) and, crucially, demonstrated the most reliable generalizability with a positive average R_{cv}^2 of 0.2604.

Although the R_{cv}^2 is lower than the overall fit due to the extreme heterogeneity of the database (variations in pipe scale, pressure, and gas species), its strictly positive value

Table 6 Comparison of model performance and parameter stability for different structures

Model structure	Flow parameter/ F^*	Overall R^2	Overall MAPE/%	Avg. validation Avg. R_{cv}^2	Max. exponent COV/%
4-Variable (Eq. 11)	$Re_g^{n_1} \left(\frac{x}{1-x} \right)^{n_2} \left(\frac{\rho_g}{\rho_f} \right)^{n_3} \left(\frac{\mu_g}{\mu_f} \right)^{n_4}$	0.9023	20.79	−0.3795 ± 2.5544	1217.16 (n_4)
3-Var-Quality-based (Eq. 16)	$Re_g^{n_1} \left(\frac{x}{1-x} \right)^{n_2} \left(\frac{\rho_g}{\rho_f} \right)^{n_3}$	0.8939	21.34	0.2604 ± 0.8740	7.54 (n_3)
3-Var-Weber-based	$Re_g^{n_1} We_f^{n_2} \left(\frac{\rho_g}{\rho_f} \right)^{n_3}$	0.8751	28.37	0.1274 ± 0.6697	6.61 (n_2)
3-Var-Froude-based	$Re_g^{n_1} Fr_f^{n_2} \left(\frac{\rho_g}{\rho_f} \right)^{n_3}$	0.8181	33.45	0.1073 ± 0.6268	12.73 (n_2)

confirms that the model is physically grounded and possesses genuine predictive power for unobserved conditions.

These results indicate that for upward annular flow, the mass quality ratio ($x/(1-x)$)—which reflects the fundamental phase distribution and interfacial shear mechanisms—acts as a more effective scaling parameter for film thickness than the direct inclusion of We_f or Fr_f . Consequently, the 3-Var-Quality-based structure was selected for the final correlation.

$$F^* = Re_g^{-0.7666} \left(\frac{x}{1-x} \right)^{-0.4806} \left(\frac{\rho_g}{\rho_f} \right)^{0.7697} \quad (16)$$

and

$$\frac{\delta}{D} = \frac{4289F^*}{1 + 42240F^*} \quad (17)$$

All exponents (n_1, n_2, n_3) in the selected 3-variable model yielded COV values below 8%, confirming that the identified power-law relationships are consistent and statistically significant across different data sources. While the LOSO-CV results reveal a relatively high uncertainty in the average R_{cv}^2 ($SD=0.8740$) and significant COV values for the leading coefficients c_1 and c_2 (50.41% and 52.63% in Table 7), this residual uncertainty is statistically expected given the model's nonlinearity and the extreme physical diversity of the 11 independent sources:

- **Fluid Property Diversity** The inclusion of non-air–water systems, such as high-viscosity air–water/glycerol (Fukano and Furukawa [12] and Alamu [24]) and low-density-ratio helium–water (MacGillivray [13]), introduces drastic shifts in interfacial momentum transfer.
- **Geometric and Operational Extremes** The database spans significant variations in scale, from standard laboratory tubes to large-diameter pipes (Gill et al. [20], $D = 31.75$ mm), and broad pressure spectrums (Sawant [21], 1.04–6.07 bar), which alter the dominance of interfacial shear versus inertial forces.

Table 7 Final identified parameters for the proposed 3-variable correlation (3-Var-Quality-based)

Parameter	Optimized value	95% Confidence interval	COV (%) [*]
c_1	4289	± 3467	50.41
c_2	42,240	$\pm 37,045$	52.63
n_1	−0.7666	± 0.0494	3.24
n_2	−0.4806	± 0.0325	4.97
n_3	0.7697	± 0.0537	7.54

^{*}COV determined from 11-fold leave-one-source-out cross-validation (LOSO-CV)

Mathematically, c_1 and c_2 function as global scaling parameters that must adapt to magnitude shifts between these distinct regimes. Therefore, their wider confidence intervals do not signify model instability, but rather reflect the necessary flexibility required to encompass a broad physical envelope within a unified correlation. The stability of the exponents serves as the primary indicator that the fundamental mechanics of film behavior remain invariant across scales and fluid properties. For a generalized framework, the current model provides the most robust and statistically sound performance.

Results and discussion

Comparison of the proposed correlation with the experimental database and existing correlations

The predictive accuracy of the developed correlation is significantly superior to existing models across the comprehensive database. This is visually demonstrated in Fig. 2, which presents the parity plots for the nine selected models. While traditional models like Henstock and Hanratty [16] or Tattersson et al. [17] show significant scattering or systematic offsets, the developed correlation exhibits a tight clustering of data points along the exact match line. As summarized in Table 3, the developed correlation achieves the highest coefficient of determination ($R^2 = 0.8939$) and the lowest global Mean Absolute Percentage Error ($MAPE = 21.34\%$). Furthermore, it provides the most robust predictions, with 77.51% of the data points falling within the $\pm 30\%$ error threshold, outperforming the next best model (Ju et al. [15] at 70.09%).

A key strength of the developed correlation is its consistent performance across diverse datasets. As detailed in Table 4, it is the only model in this study to maintain a source-wise MAPE of less than 20% for nearly all air–water datasets except the large-diameter pipe experiments of Gill et al. [20]. However, the analysis identifies specific challenges when predicting datasets involving non-aqueous liquids, such as Alamu [24] and the glycol–water mixtures of Fukano and Furukawa [12].

Despite its global performance, the model exhibits limitations when applied to non-aqueous fluids (glycerol/water mixtures). According to Table 4, the model yields source-wise MAPEs of 61.16% for Alamu [24] and 42.65% for the air–water/glycol data of Fukano and Furukawa [12]. Nevertheless, the developed correlation demonstrates superior stability in these challenging cases; it stands as one of only two models, alongside Berna et al. [14], that avoids extreme error divergence across both glycol-based datasets. In contrast,

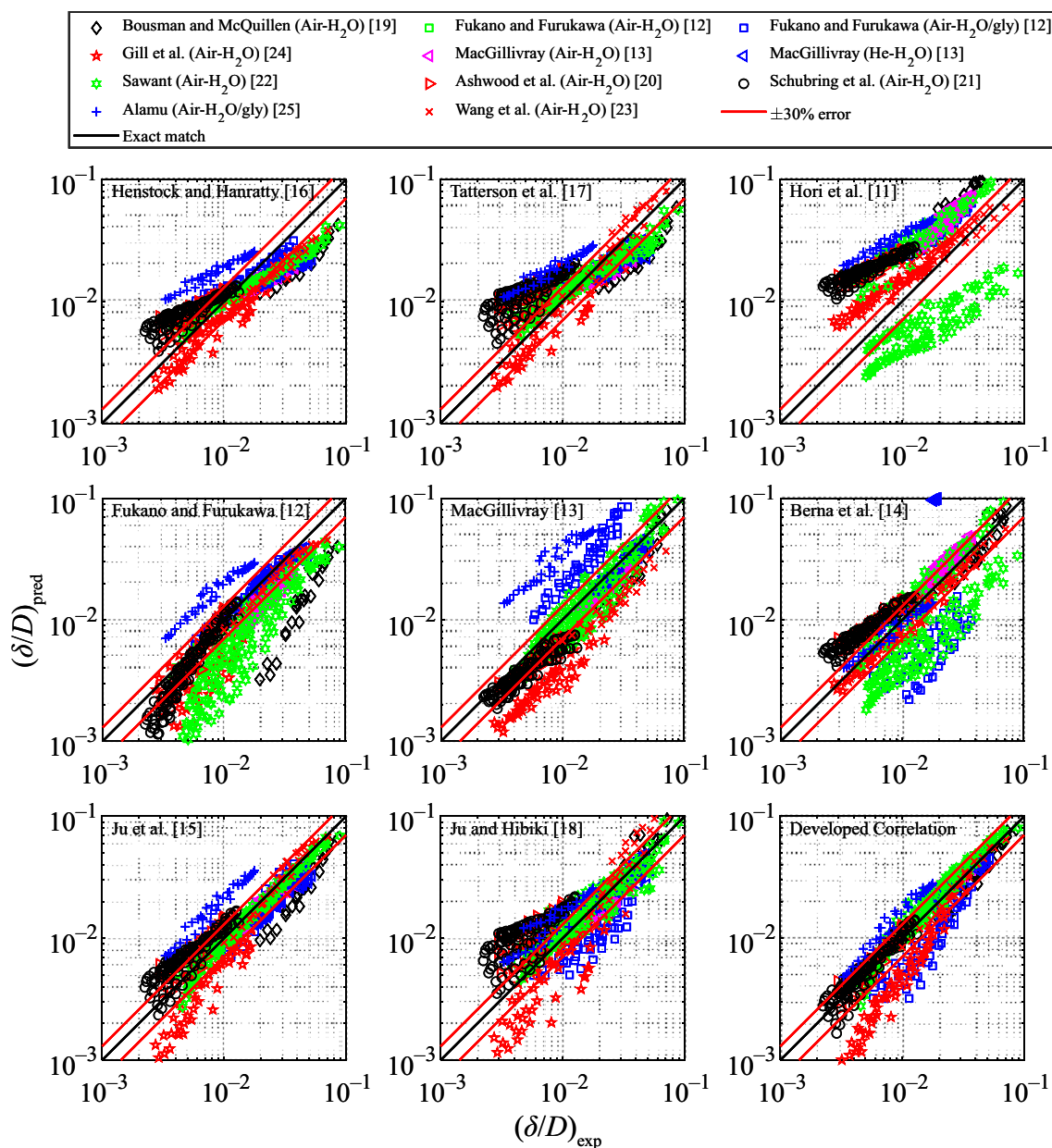


Fig. 2 Comparison of experimental and predicted dimensionless liquid film thickness (δ/D) for various predictive models. The solid black line represents the exact match, and the red lines indicate the $\pm 30\%$ error bounds

other established models often succeed with one dataset but fail catastrophically with the other. For instance, the models of Fukano and Furukawa [12] and Ju et al. [15] provide highly accurate predictions for the air–water/glycerol data of Fukano and Furukawa [12] (7.88% and 12.45% MAPE, respectively) but totally fail when applied to Alamu’s [24] data, with errors surging to 98.88% and 112.41%. While Berna et al. [14] achieves a lower error on Alamu’s [24] data (12.49% MAPE), its performance on Fukano and Furukawa’s [12] glycol data still results in a significant MAPE of 63.42%. These results indicate that the developed correlation

offers a more balanced and reliable prediction for high-viscosity fluids across different experimental sources, even where other sophisticated models lack generalizability.

Intriguingly, the parity plots in Fig. 2 reveal a lack of a singular systematic trend for these high-viscosity fluids. The model exhibits discrepancies in two distinct ways. It tends to overpredict the liquid film thickness for the glycol–water data reported by Alamu [24], while underpredicting the glycol–water data of Fukano and Furukawa [12]. This inconsistency suggests that the physical mechanisms governed by viscosity and surface tension are highly complex and may

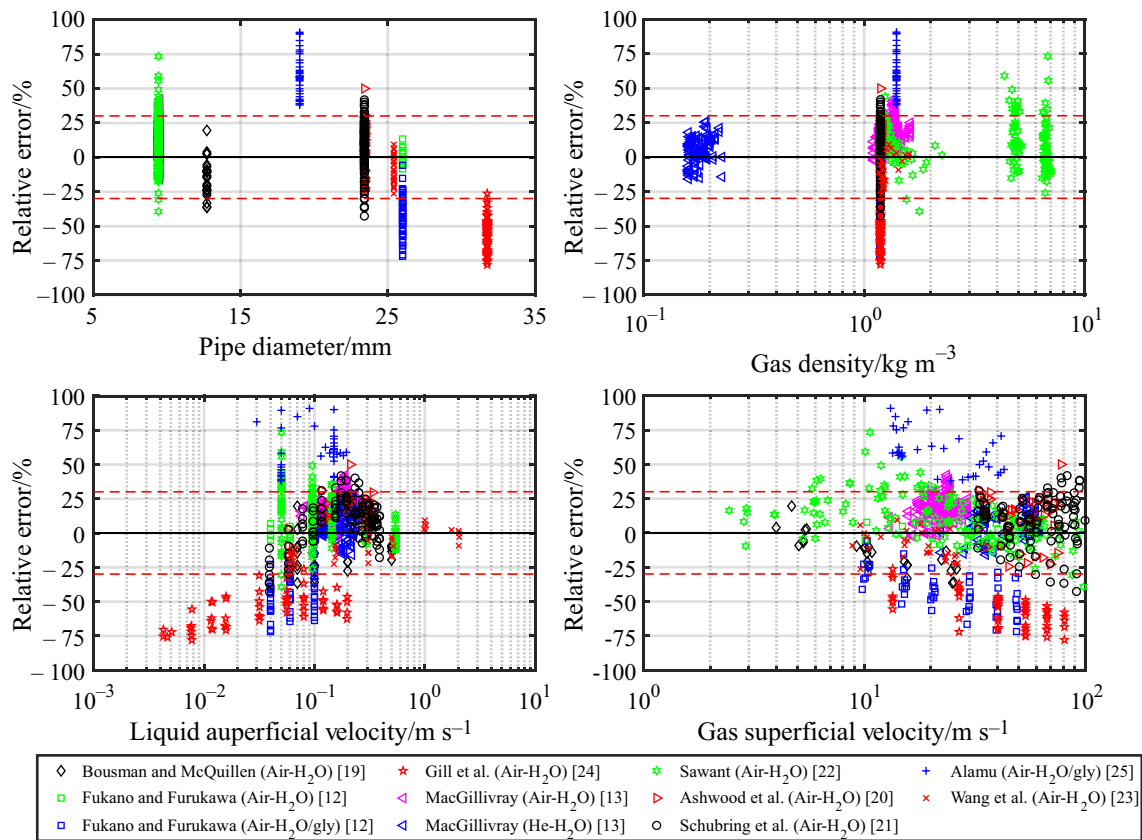


Fig. 3 Relative error distribution of the developed correlation against key operating parameters: pipe diameter, gas density, liquid superficial velocity, and gas superficial velocity. The dashed red lines represent the $\pm 30\%$ error thresholds

not yet fully capture the nonlinear interaction between liquid viscosity and interfacial shear.

These discrepancies may also be attributed to differences in experimental techniques and measurement instruments. For example, Alamu [24] employed a conductance probe to measure the film thickness of highly viscous glycerol–water mixtures, whereas most other studies primarily utilized optical techniques, such as planar laser-induced fluorescence (PLIF) or total internal reflection (TIR). Electrical and optical measurement methods demonstrate varying sensitivities to interfacial wave structures and film surface fluctuations. These findings highlight that, while the dimensionless parameters in the newly developed model are more physically justified than those in previous studies, additional validation with diverse working fluids and rigorously chosen measurement techniques is required to more precisely isolate and elucidate the underlying physical mechanisms.

To provide deeper insight into the model's robustness, Fig. 3 presents relative error distribution plots against pipe diameter, superficial velocities, and gas density, which are the primary physical and geometric variables governing interfacial dynamics in annular flow. The pipe diameter (D) is selected to confirm the scalability of the dimensionless

structure (δ/D) across varying geometric scales, while the superficial velocities (j_g, j_f) are included as the drivers of momentum and interfacial shear stress, helping to identify model limitations across flow regimes where surface tension effects may become significant. Notably, gas density (ρ_g) is utilized instead of pressure to provide a more fundamental physical analysis of gas-phase inertia; since the database incorporates both Air and Helium, density accounts for both pressure and molecular weight, effectively identifying systematic biases across different gas species that might otherwise be obscured.

As illustrated in the error distribution plots (Fig. 3), the model demonstrates no significant systematic bias relative to pipe diameter for the majority of the database, covering diameters up to 26 mm, with errors remaining largely within the $\pm 30\%$ bounds. However, the dataset from Gill et al. [20] stands out due to its larger diameter of 31.75 mm, where a systematic underprediction (MAPE of 55.90%) is observed. This suggests a potential scale effect where gravitational forces and film distribution around the periphery differ from smaller-scale data. This geometric disparity, combined with the liquid superficial velocity (j_f) analysis, reveals significant fluctuations in the low-velocity regime ($j_f < 0.1 \text{ m s}^{-1}$).

Such deviations indicate that at lower liquid flow rates and larger physical scales, surface tension effects—which govern interfacial wave stability—play a more dominant role than currently accounted for in the model's inertial terms.

In contrast, the model exhibits high stability across the range of gas superficial velocities (j_g) and gas densities (ρ_g) without any clear systematic bias. Notably, the correlation shows exceptional accuracy when tested against very low-density gases, such as the Helium–water data from MacGillivray [13], with error values clustered closely around the 0% line throughout the density range. This comprehensive approach ensures a physically meaningful and robust evaluation of the model's accuracy across a wide range of fluid properties and experimental conditions.

Another important observation is that empirical and semi-empirical models tend to perform well when applied to datasets used in their development, but may have limitations when tested against entirely independent datasets. The new correlation, despite being based on a semi-empirical approach, follows a more generalized framework, enabling it to capture the film thickness behavior across multiple datasets more effectively than purely empirical models. However, variations in experimental setups, measurement techniques, and test conditions can introduce uncertainties that influence model accuracy.

From a practical standpoint, the explicit formulation of the developed correlation enhances its usability in engineering applications, especially for predictive design in heat exchangers, boiling and condensation systems, and two-phase flow simulations. The model serves as a valuable tool for improving thermal management and process efficiency in industrial applications by reducing uncertainty in liquid film thickness predictions.

Added value compared with existing models

Although the dimensionless groups employed in this study have individually appeared in prior works, their integration into a newly structured flow parameter provides a distinct balance between physical interpretability and empirical flexibility. This differs from existing correlations, which either rely on complex mechanistic formulations (Henstock and Hanratty [16]; Ju and Hibiki [18]) or are tailored to limited datasets (Fukano and Furukawa [12]; MacGillivray [13]).

As summarized in Table 3, the proposed correlation achieves the lowest overall mean absolute percentage error (21.34%) and the highest proportion of predictions within 30% error bounds (up to 77.51%) across the full database of 876 points. This demonstrates improved generalizability compared with models that perform well only for the datasets on which they were originally developed.

While mechanistic models such as that of Alves et al. [10] can, in principle, predict annular film thickness from

momentum balance equations, they require detailed knowledge of interfacial and wall shear stresses that are rarely available in practical applications. Their implicit formulations also increase implementation complexity. In contrast, the present semi-empirical correlation provides an explicit and user-friendly closed-form equation, representing a balanced compromise between theoretical foundations and practical applicability. This combination of accuracy, simplicity, and physical interpretability constitutes a key strength of the proposed model for engineering energy-system design applications.

Conclusions

A comprehensive database of liquid film thickness in upward vertical annular gas–liquid flow was compiled from nine experimental studies. Building upon this dataset, a new explicit semi-empirical correlation was developed that preserves the underlying physical mechanisms of annular flow while overcoming the limitations of previous empirical models.

The proposed correlation is derived from the shear stress equations of a fully developed wall layer and verified using extracted data specifically targeted at fully developed flow conditions. The finalized model relies on three dimensionless groups: the gas Reynolds number, the gas-to-liquid mass flux ratio, and the gas-to-liquid density ratio.

The applicable range of the proposed correlation is defined by the following bounds:

- Flow State: Fully developed upward vertical annular flow
- Tube Diameter (D): 9.40 to 26 mm
- System Pressure: Ambient (~ 1.0 bar) up to 6.07 bar
- Liquid Superficial Velocity (j_f): 0.04 to 2.049 m s⁻¹
- Gas Superficial Velocity (j_g): 2.44 to 114.30 m s⁻¹

The proposed correlation significantly outperforms the existing models evaluated in this study, achieving a mean absolute percentage error (MAPE) of 21.34% and a coefficient of determination (R^2) of 0.8939. In addition, 77.51% of the predictions fall within a $\pm 30\%$ error range.

The correlation performs well and is suitable for energy-related systems. However, it has limitations with non-water and highly viscous fluids, such as glycerol–water solutions, which can lead to greater prediction deviations. Future research should expand the experimental database to include more high-viscosity fluids and address uncertainties from different measurement techniques. Incorporating advanced turbulence modeling and machine learning may further enhance predictive accuracy. Despite these limitations, the correlation remains an effective and practical tool for engineering applications involving annular

two-phase flow, especially in energy-related systems such as nuclear power plant cooling, advanced heat exchangers, and chemical processing equipment.

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Data availability No datasets were generated or analyzed during the current study.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT in order to improve the writing and English of some sections. After using this tool/service, the authors reviewed and edited the content as needed and they take full responsibility for the content of the publication.

Declarations

Competing interests The authors declare no competing interests.

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